

# **Linear Regression Prediction of Hospital Stay Duration**

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## **Motivation**

I volunteer at the Johns Hopkins Hospital, which recently stopped accepting United Healthcare insurance. It was a stark wake-up call about the costs of healthcare in the United States – costs that increase significantly with longer stays.

At the same time, I noticed units without empty beds. Just as long hospital stays cost more for patients, they pose an additional burden to hospital logistics staff. If we have a way of predicting how long a patient will spend in the hospital, the patient can plan their finances accordingly- and hospitals can get a better sense of where to place their patients (including patients to different rooms as need be).

## **Background**

The average hospital stay (as of 2018) lasted five and a half days; overly-extended hospital stays negatively affect both patients and hospital staff, potentially even leading to worsened patient outcomes (Tipton et al., 2021). Additionally, hospital bed shortages are projected to increase – especially after the extreme shortages seen during the COVID-19 pandemic (Leuchter et al., 2025). When a hospital lacks the bed space for a patient, that patient is kept waiting in the emergency room, transported to another hospital (that may be farther from home or offer a lower standard of care), or – in the worst case scenario – denied care altogether.

At the same time, healthcare costs (especially in the US) are practically higher than ever, and these costs increase exponentially per day that a patient is hospitalized (McAllister, 2024). Doctors have recently also been under increased pressure from health insurance companies to reduce “nonessential” costs, placing additional burdens on the healthcare teams (and, again, ultimately impacting patients) (Richards et al., 2022).

Needless to say, this is not a good situation. I thought it would be interesting, and theoretically beneficial to hospitals, to model how long a patient is expected to stay in a hospital, based on their current health status and other factors. With a system like this, hospitals will be able to better allocate rooms (preventing backup and overflow issues), patients will have an increased ability to properly prepare for their hospital stays, and insurance companies will (in an ideal universe) stop pressuring doctors to move patients in and out of the hospital as quickly. This is a healthcare center, people – not a fast food joint.

## Process

Web scraping was beyond the limits of this class (although I guess I could have asked an LLM for help... but I digress), so I first downloaded theoretical patient information from Kaggle. I tried a few different datasets, and I think it's worth mentioning that the first one I tried included patient names! My first step there was to delete said names and not risk my nonexistent medical license on a HIPAA violation. This is the data set I decided on using:

<https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset>

As with the homework assignment that dealt with a similar topic, my first step was to prepare the data for processing. I got rid of two columns ("noise" and "notes") that were irrelevant to the topic at hand. Some columns also had missing data (\*gasp\*), so the blanks were substituted with the median values from the columns. This made it such that I did not have to delete the 4500 entries with missing data. Medians, specifically, were chosen over averages or zeroes because they are the least susceptible to being skewed by extreme values.

Once everything was prepared, a correlation matrix was used to evaluate which factors would have the greatest impact on patient stay. A linear relationship was denoted with these factors, which made a logistic model a good candidate for this project!

The model was applied and trained on training data, then evaluated on testing data.

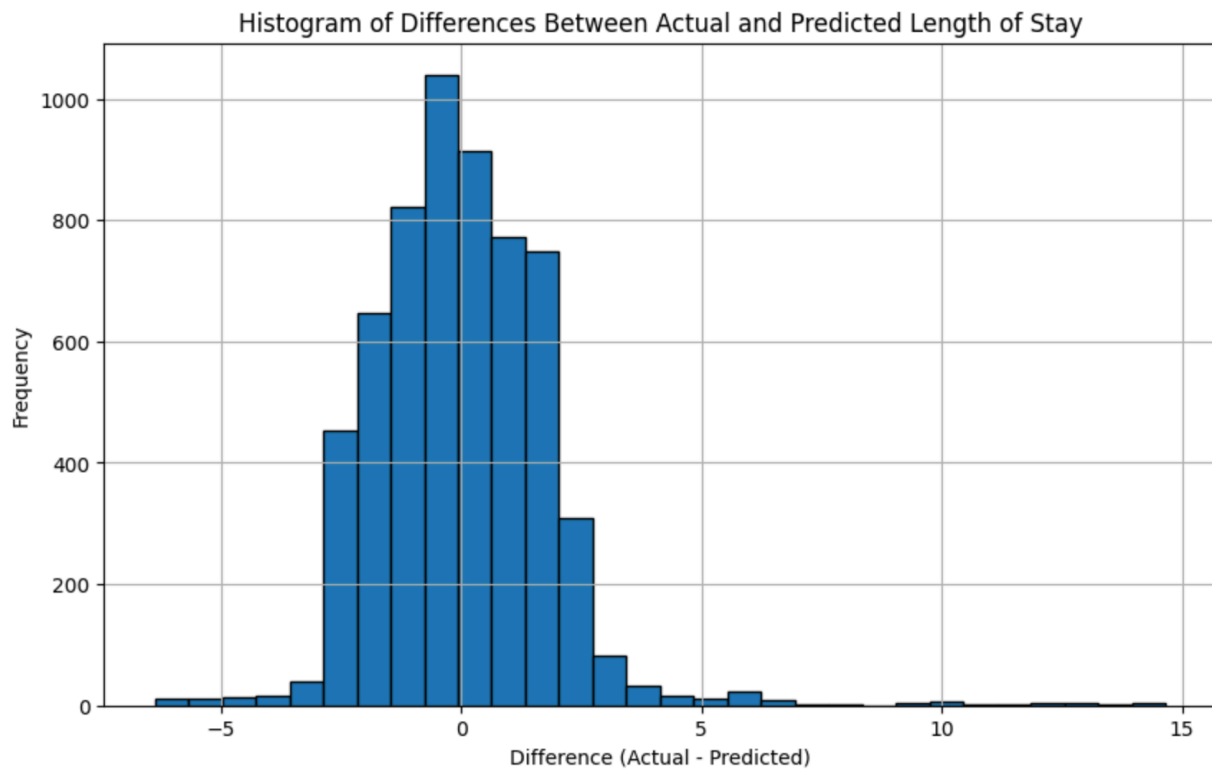
## Results

As seen in **Figure 1**, the average difference between what my model predicted and the actual length of stay was relatively low, which is exciting. There was a good bit of variation, as well as some significant outliers, though (as shown in the relatively high standard deviation and min & max difference values). **Figure 2** shows a histogram of these difference values.

The Mean Squared Error (MSE) calculated was  $\sim 3.26$ , and the R-squared ( $R^2$ ) Score was  $\sim 0.58$ . These values indicate a moderate "success rate," which is not terrible when considering that a linear regression model was applied to pretty complicated data.

| LengthOfStay |             |
|--------------|-------------|
| count        | 6000.000000 |
| mean         | 0.000324    |
| std          | 1.806384    |
| min          | -6.377348   |
| 25%          | -1.263999   |
| 50%          | -0.116218   |
| 75%          | 1.048218    |
| max          | 14.658025   |

**Figure 1** - “Basic statistics” of deviation between predicted and actual stay lengths on test data



**Figure 2** - Histogram of difference values (between predicted and actual stay length)

## Limitations of Current Analysis & Future Directions

This analysis was conducted on relatively “unfinished” data uploaded by a Kaggle user and acquired online for free. It is, obviously, of relatively low quality. Finally, much of the data was filled in from medians and modes, making for significantly less reliable data. This was done so that I would be able to work with the data, but it significantly reduced the quality of said data. To account for this, a good future direction would be to apply my model to better data. This data could be acquired via web scraping, or even by manual collection at the Johns Hopkins Hospital!

I also have to admit that a linear regression analysis is relatively crude for the complexity of the data. Other regression models, like polynomial or ridge regression, would do a better job accounting for non-linear relationships and other hidden complexities in the data selected. A more advanced neural network would likely do an even better job of analyzing this data! Attempting these models would also be a strong future step.

## Sources

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